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C 6. A 5 m and 15 m long aluminum rods are heated from 20°C to 80°C. How do you compare the coefficient of linear expansion of the two rods?

- The longer rod has a greater coefficient of linear expansion.
 - The shorter rod has a greater coefficient of linear expansion.
 - Both rods have the same coefficient of linear expansion.
 - The coefficient of linear expansion of the longer rod is exactly three times the shorter rod.
- A 7. A water stream narrows as it falls (i.e. $A_b > A_t$) as shown in the figure. Compare the speed of the water at these two points (A_t and A_b) after the water falls height h down. (Neglect viscosity) Where V_t : velocity at A_t and V_b : velocity at A_b .

- $V_b < V_t$
- $V_b = V_t$
- $V_b > V_t$
- $V_t = h$



B 8. A 0.5 mole of gas at temperature of 300 K expands isothermally from an initial volume of 3 L to 6 L. The work done by the gas and the possible added heat to the gas respectively are: ($R = 8.31 \text{ J/mol K}$; $\ln(2) = 0.693$; $\ln(0.5) = -0.693$)

- 864 J, -864 J
- 864 J, 864 J
- 1728 J, -1728 J
- 1728 J, 1728 J

D 9. Which one of the following is NOT the criteria for an oscillation to be said simple harmonic motion?

- The force is directed towards the equilibrium position.
- The system is described by sine and cosine functions.
- The force is proportional to the displacement.
- The direction of the force is similar to the direction of the displacement.

A 10. A simple pendulum of length 80 cm is oscillating in small angle about an equilibrium position with a period of 2 s. what is the acceleration due to gravity at this position?

- 7.89 m/s^2
- 9.81 m/s^2
- 11.5 m/s^2
- 16.1 m/s^2

DIRECTION II: Give short answers that best fill the blank spaces following each question.(3 pos)

- Latent heat is the thermal energy supplied to overcome the force of attraction between the molecules of a substance during a phase change.
- A particle is in simple harmonic motion along the x axis. The amplitude of the motion is 4. At one point in its motion its kinetic energy is $KE = 5 \text{ J}$ and its potential energy is $U = 3 \text{ J}$. When it is at $x = 4$, the kinetic and potential energies are: $KE = 0 \text{ J}$ and $U = 7 \text{ J}$.
- Let the temperature of a body is raised by 50°C. What will be this change in the temperature on the Fahrenheit and Kelvin scales? 112 °F and 113.6 K
- The temperatures of equal masses of three different liquids A, B and C are 12°C, 18°C and 28°C respectively. When A and B are mixed the temperature becomes 16 °C and when B and C are mixed, it is 23 °C. The temperature of the mixture of A and C is 15.5 °C.
- For an oscillator performing simple harmonic motion, fill the following table with appropriate value (equation or zero). (Use $\pm$ for equations having positive and negative value with time).

	velocity	acceleration	Kinetic energy	Potential energy
At equilibrium	$\pm \omega A$	0	$\frac{1}{2} k A^2$	0
At maximum displacement	0	$\pm \omega^2 A$	0	$\frac{1}{2} k A^2$



- Switch
- Cheating and attempt to cheat will disqualify
- Attempt all the questions as intended!
- This question paper contains 10 multiple choices, 5 short answer and 3 w questions confined in 5 pages. Check all the questions are avail before starting attempt.

DIRECTION I: Choose the correct answer from the given alternatives and write the letter choice in CAPITAL LETTER on the space provided in front of each respective question.

- A 1. The radius of a water pipe decreases from 10 to 5 cm. If the average velocity in the is 4 m/s, find the average velocity in the narrower region.
 A. 16 m/s B. 8 m/s C. 20 m/s D. 4 m/s
- D 2. A thin convex lens has a focal length of 25.0 cm. What would be the position of the f when the object was placed 50.0 cm in front of the lens?
 A. 35.0 cm B. infinity C. 25.0 cm D. 50.0 cm
- C 3. Let an oscillator is oscillating about an equilibrium position. If the oscillator is at $x = A$ position of the particle at any time t can be expressed by:
 A. $X(t) = A \sin \omega t$ B. $X(t) = \omega A \cos \omega t$ C. $X(t) = A \cos \omega t$ D. $X(t) = \omega A \sin \omega t$
- D 4. Which one of the following is TRUE about wavelength and frequency when listener?
 A. Both increase
 B. Both decrease
 C. Wavelength increase
 D. Wavelength decrease
- A 5. A bimetallic strip made up of iron and copper is heated as shown in the figure expansion of copper is smaller than that of iron. What result do you expect?
 A. The strip bends upward
 B. The strip bends down ward
 C. The strip elongates without bending
 D. The strip contracts without bending





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Multiple Choice Questions (38 Points)

I. PRELIMINARIES (1 pts)

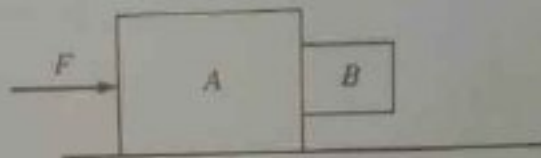
1. Two vectors are given by $\vec{A} = 4\vec{i} - 3\vec{j} + \vec{k}$ and $\vec{B} = -\vec{i} + \vec{j} + 4\vec{k}$. Find a vector $\vec{C}$ such that $\vec{A} - \vec{B} + \vec{C} = 0$

- A. $-4\vec{i} + 3\vec{j} - \vec{k}$
- B. $-5\vec{i} + 4\vec{j} + 3\vec{k}$
- C. $3\vec{i} - 2\vec{j} + 5\vec{k}$
- D. $5\vec{i} - 4\vec{j} - 3\vec{k}$
- E. $\vec{i} - \vec{j} - 4\vec{k}$

II. KINEMATICS AND DYNAMICS OF PARTICLES (5 pts)

1. For the system consisting of the two blocks shown in the figure, the minimum horizontal force F is applied so that block B does not fall under the influence of gravity. The masses of A and B are 16.0 kilograms and 4.00 kilograms, respectively. The horizontal surface is frictionless and the coefficient of friction between the two blocks is 0.50. The magnitude of F is most nearly

- A. 50 N
- B. 100 N
- C. 200 N
- D. 400 N
- E. 1,600 N



2. Just as a car starts to accelerate from its initial velocity 4 m/s with acceleration 1.4 m/s², a bus moving with constant speed of 16 m/s passes it in a parallel lane. How long does it take for the car to overtake the bus?

- A. 17.1 s
- B. 8.6 s
- C. 2.9 s
- D. 14.3 s
- E. It will never overtake the bus

3. A cannonball is fired with a velocity v . At what angle from the ground must the cannonball be fired in order for it to hit an enemy that is at the same elevation, but a distance d away?

- A. $\sin^{-1}(v/gd)$
- B. $\sin^{-1}(gd/2v)$
- C. $\sin^{-1}(2gd/v)$
- D. $(1/2)\sin^{-1}(gd/v^2)$
- E. $\sin^{-1}(gd/v^2)$

4. In one-dimensional collision, a particle of mass $2m$ collides with a particle of mass m at rest. If the particles stick together after the collision, what fraction of the initial kinetic energy is lost in the collision?

- A. 0
- B. 1/4
- C. 1/3
- D. 1/2
- E. 2/3

5. A satellite of mass m orbits a planet of mass M in a circular orbit of radius R . The time required for one revolution is

A. independent of M
 B. proportional to $\sqrt{m}$
 C. linear in R
 D. proportional to $R^{3/2}$
 E. proportional to R^2

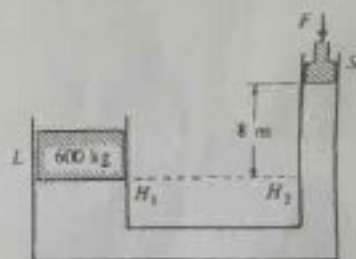
III. FLUID MECHANICS (12 pts)

6. Find the ratio of a systolic blood pressure of 120 (in mmHg) to atmospheric pressure. Standard atmospheric pressure is $1.01 \times 10^5 \text{ Pa}$ and density of mercury is 13.6 g/cm^3

A. 1.00
 B. 1.16
 C. 0.16
 D. 0.84
 E. 6.31

7. For the hydraulic system shown in the figure, the cylinder on the left has a mass of 600 kg and cross-sectional area of 800 cm^2 . The piston on the right has cross-sectional area 25 cm^2 and negligible weight. If the apparatus is filled with oil ($\rho = 0.78 \text{ g/cm}^3$), what is the force F required to hold the mass in equilibrium?

A. 187.5 N
 B. 62.4 N
 C. 209.8 N
 D. 125.1 N
 E. 31 N



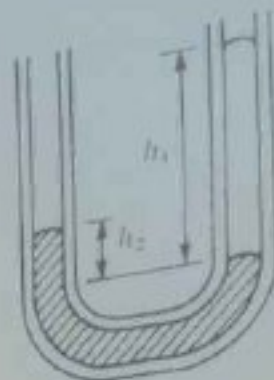
8. A diver in water picks up a lead cube of side length 10 cm. How much force is needed to lift the cube? The density of lead is approximately 11 g/cm^3 .

A. 10 N
 B. 11 N
 C. 50 N
 D. 100 N
 E. 110 N

9. A boat floating in fresh water displaces 3.63 m^3 of water. What volume of water would this boat displace if it were floating in salt water of density 1024 kg/m^3

A. 3.55 m^3
 B. 3.63 m^3
 C. 3.72 m^3
 D. 7.17 m^3
 E. 0.087 m^3

15. Assume that the two liquids in the U-shaped tube, shown in the figure, are water and oil. Compute the density of the oil if the water stands 19 cm above the interface and the oil stands 24 cm above the interface. Density of water is 1000 kg/m^3



- A. 1236 kg/m^3
- B. 5000 kg/m^3
- C. 792 kg/m^3
- D. 208 kg/m^3
- E. 1000 kg/m^3

16. An aqueduct consisting of a pipe filled completely with water passes up a hill that is 10 m high. At the bottom of the hill, a flowmeter measures the speed of the fluid to be 2.0 m/s. At the top of the hill, a flowmeter measures the speed of the fluid to be 1.0 m/s. Which is the closest to the difference in the fluid pressure between the bottom and top of the hill?

- A. $1.50 \times 10^3 \text{ Pa}$
- B. $9.85 \times 10^4 \text{ Pa}$
- C. $1.00 \times 10^5 \text{ Pa}$
- D. $1.02 \times 10^5 \text{ Pa}$
- E. $9.85 \times 10^7 \text{ Pa}$

17. An incompressible fluid of density ρ flows through a horizontal pipe of radius r and then passes through a constriction of radius $r/2$. If the fluid has pressure P_0 and velocity v_0 before the constriction, the pressure in the constriction is

- A. $P_0 - (15\rho v_0^2/2)$
- B. $P_0 - (3\rho v_0^2/2)$
- C. $P_0/4$
- D. $P_0 + (3\rho v_0^2/2)$
- E. $P_0 + (15\rho v_0^2/2)$

IV. HEAT AND THERMODYNAMICS (20 pts)

18. Why a balloon filled with hot air floats

- A. The air in balloons expands and becomes more dense than the surrounding air
- B. The air in balloons expands and becomes less dense than the surrounding air
- C. The air in balloons has no difference in density with the surrounding air
- D. None of the above

19. In thermodynamics, the internal energy of any system includes

- A. kinetic energy of translation of molecules
- B. rotation of molecules
- C. potential energy of molecules
- D. All of the above

32. Body A has twice the mass and three times the specific heat of body B. They are supplied with equal amounts of heat. Body A experiences a temperature change ΔT . What change in temperature is experienced by body B?

A. ΔT

C. $2\Delta T/3$

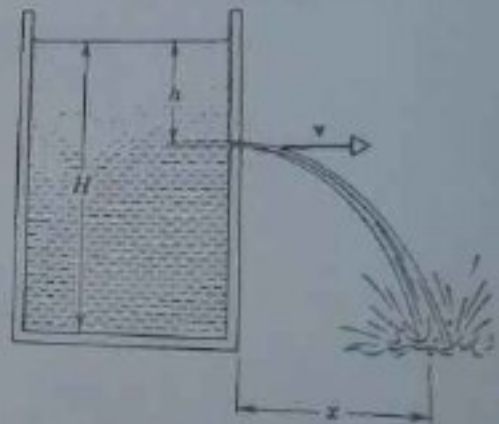
B. $3\Delta T/2$

D. $6\Delta T$

Workout Problems (12 Pts)

1. A ball of mass m is launched at 45° from the horizontal by a spring of spring constant k that is compressed by a displacement d . Show that the horizontal distance x the ball travels before landing on the ground is given by kd^2/gm .

2. A tank is filled with water to a height H . A hole is punched in one of the walls at a depth h below the water surface, as shown in the figure. Show that the horizontal distance x from the edge of the bath where the draining water land is given by $2\sqrt{h(H-h)}$.



Booklet Code: 1101-D04

Phys1011 Final Exam for First Year Students, 2020, UoG. Booklet Code: 1101-D04

3. A 0.050 kg bar of metal is heated to 200.0°C and then immersed into a beaker containing 0.400 kg of water initially at 20.0°C . If the final equilibrium temperature of the mixed system is 22.4°C , find the specific heat of the metal (Take Specific heat of water as $4186 \text{ J/kg } ^{\circ}\text{C}$)



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$$0.4 \times 10^{-3} \text{ kg/m}^3 \times 12 \times 10^{-4} \text{ m}^3$$

$$\boxed{4.8 \times 10^{-2} \text{ kg}}$$

$$mg = \boxed{4.8 \times 10^{-3} \text{ kg}}$$

$$V = \frac{4}{3} \pi (0.05)^3 = \frac{4}{3} \pi \times 125 \times 10^{-6}$$

$$1000 \text{ kg/m}^3 \times 125 \times 10^{-6} \text{ m}^3 \times 10 = 12\pi \times 10^{-4} \text{ m}^3$$

Part I. Multiple choice questions (1.5pt each)

- Which statement is correct about the pressure under a fluid in a container?
 - It depends on the volume of the fluid
 - It depends on the shape of the container
 - It depends on the density of the fluid
 - It is independent of the depth of the fluid
- A table-tennis ball which is spherical in shape has a diameter of 6 cm and average density of 0.4 g/cm^3 . What force is required to hold it completely submerged under water?
 - 0.68 N downward
 - 0.68 N upward
 - 1.13 N downward
 - 1.13 N upward
- Lead has a greater density than iron, and both are denser than water. Comparing the buoyant force on the lead object to the buoyant force on the iron object of same volume,
 - the buoyant force on the lead object is greater than the buoyant force on an iron object
 - the buoyant force on the lead object is less than the buoyant force on an iron object
 - the buoyant force on the lead object is equal to the buoyant force on an iron object
 - None of the above
- Two tubes of different cross-sectional areas are connected together horizontally. Which of the following is a true statement about a fluid flowing through the tubes
 - The amount of fluid that leaves the wider tube is less than the amount of fluid that enters the narrower tube in the same interval of time.
 - The speed of the fluid in the narrower tube is greater than the speed of the fluid in the wider tube.
 - The pressure of the fluid in the narrower tube is greater than that in the wider tube.
 - The speed of the fluid in the narrower tube is less than the speed of the fluid in the wider tube.
 - The amount of fluid that leaves the wider tube is greater than the amount of fluid that enters the narrower tube in the same interval of time.
- A garden hose has an inside diameter of 16 cm. The hose can fill a 10 liter bucket in 20 seconds. If the diameter of the nozzle is reduced to 8 cm,
 - Water exits with a speed twice the speed inside the hose.
 - A 10 liter bucket can be filled in 10 seconds
 - A 20 liter bucket can be filled in 40 seconds
 - Water exits with a speed half the speed inside the hose.

$$V = 36\pi \times 10^{-6} \text{ m}^3$$

$$\rho = 4 \times 10^3 \text{ kg/m}^3$$

$$m = V\rho = 0.04522 \text{ kg}$$

$$V = 0.4 \times 10^{-3} \text{ kg} \times 36\pi \times 10^{-6} \text{ m}^3$$

$$0.4 \times 36\pi \times 10^{-9}$$

$$45.216 \times 10^{-9} \text{ kg}$$

$$V = \frac{4}{3} \pi r^3 = \frac{4}{3} \pi (27 \times 10^{-6} \text{ m}^3)$$

$$\boxed{36\pi \times 10^{-6} \text{ m}^3}$$

6. When a load of 500 kg is hanging from a steel wire of length 3 m and cross sectional area 0.2 cm^2 , the wire stretches beyond its no load length. If the young's modulus for the wire made of steel is $Y = 2 \times 10^{11} \text{ Pa}$, what is the change in length of the wire beyond its no-load length?

- (A) 3.75 mm (B) 2.75 mm (C) 1.75 mm (D) 0.75 mm

7. What is Thermal expansion?

A) The transfer of thermal energy between materials by the collisions of particles

B) Is a decrease in a materials size when temperature decreases

C) An increase in a materials size when temperature increases

(D) All

$$Y = \frac{F}{A} \frac{\Delta l}{l_0}$$

$$Y = \frac{F l_0}{A \Delta l}$$

$$\Delta l = \frac{F l_0}{Y A}$$

8. Which of the following statements is false?

A) Gaps need to be left in lengths of railway lines to prevent buckling in hot weather

B) Bimetallic strips are used in thermostats, a thermostat being a temperature- operated switch

C) As the temperature of water is decreased from 4°C to 0°C contraction occurs

D) A change of temperature of 15°C is equivalent to a change of temperature of 15 K

9. If the coefficient of linear expansion is A, the coefficient of superficial expansion is B and the coefficient of cubic expansion is C, which of the following is false?

A. $C = 3A$

B. $A = B/2$

C. $B = 3C/2$

D. $A = C/3$

10. The temperature of land rises more quickly than that of the sea because the specific heat of soil is

A) more than water

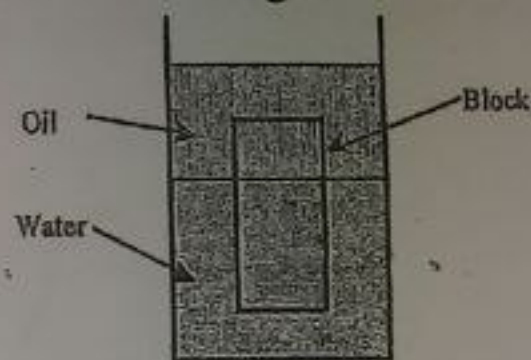
B) equal to water

C) less than water

D) neutral

Part II. Give short answer /1.5pt each/

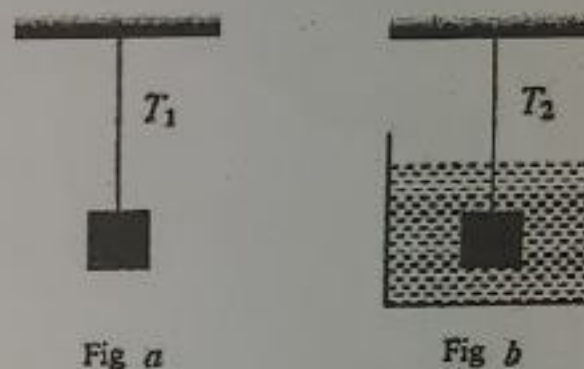
1. A layer of oil with density 800 kg/m^3 floats on top of a volume of fresh water. A block of some material of density 950 kg/m^3 , a uniform cross-sectional area A and height h floats at oil-water interface, as shown in figure below.



- (a) What fraction of the block is in water? _____
 (b) What fraction of the block is in oil? _____

2. An object is suspended by a cord as shown in *figure 'a'* and the tension in the cord is $T_1 = 10 \text{ N}$. The object is then immersed to the water as shown in *figure 'b'*. In this case the tension in the cord is $T_2 = 6 \text{ N}$.

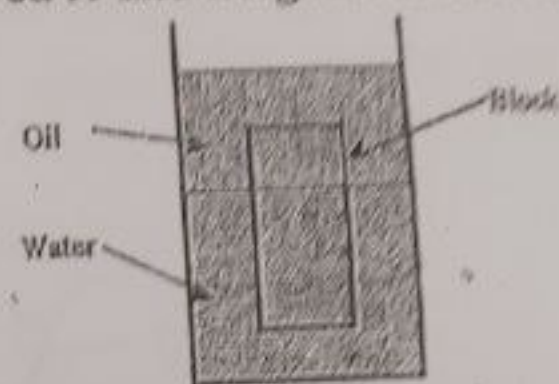
Determine the density of the object. _____



3. A small block of wood, of density $0.4 \times 10^3 \text{ kg/m}^3$, is submerged in water at a depth of 2.9 m . Find
- (a) The acceleration of the block towards the surface when the block is released. _____
- (b) The time for the block to reach the surface. _____
4. If a pendulum clock keeps perfect time at the base of a mountain, will it also keep perfect time when it is moved to the top of the mountain? Explain. _____
5. A mass-spring system moving with simple harmonic motion has amplitude A . When the kinetic energy of the object equals twice the potential energy stored in the spring, what is the position x of the object? _____
6. If the spring having spring constant k is cut in to two equal parts, the spring constant of either half will be _____

Part II. Give short answer /1.5pt each/

1. A layer of oil with density 800 kg/m^3 floats on top of a volume of fresh water. A block of some material of density 950 kg/m^3 , a uniform cross-sectional area A and height h floats at oil-water interface, as shown in figure below.



- (a) What fraction of the block is in water? _____
 (b) What fraction of the block is in oil? _____

2. An object is suspended by a cord as shown in *figure 'a'* and the tension in the cord is $T_1 = 10 \text{ N}$. The object is then immersed to the water as shown in *figure 'b'*. In this case the tension in the cord is $T_2 = 6 \text{ N}$.

Determine the density of the object. _____

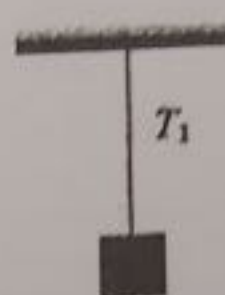


Fig a

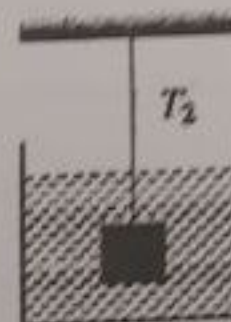


Fig b

3. A small block of wood, of density $0.4 \times 10^3 \text{ kg/m}^3$, is submerged in water at a depth of 2.9 m . Find

(a) The acceleration of the block towards the surface when the block is released. = _____

(b) The time for the block to reach the surface. _____

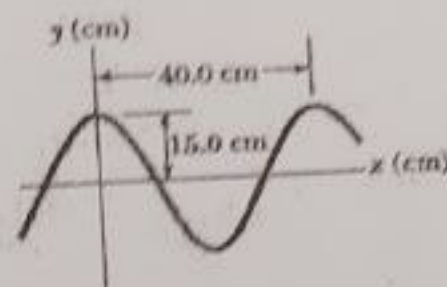
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5. A mass-spring system moving with simple harmonic motion has amplitude A . When the kinetic energy of the object equals twice the potential energy stored in the spring, what is the position x of the object? _____

6. If the spring having spring constant k is cut in to two equal parts, the spring constant of either half will be _____

7. The condition of resonance occurs when the driving frequency becomes equal to _____ of the oscillator.

8. A sinusoidal wave traveling in the positive x-direction has a wavelength of 40cm, and a frequency of 8Hz. The vertical position of an element of the medium at $t = 0$ and $x = 0$ is also 15cm as shown in figure below. The period T & linear speed of the wave are _____ and _____ respectively.



9. A car with a mass of 1300 kg is constructed so that its frame is supported by four springs. Each spring has a force constant of 20,000 N/m. Two people riding in the car have a combined mass of 160 kg. Find the frequency of vibration of the car after it is driven in the road and the car oscillates vertically. _____

10. Rubber has a negative average coefficient of linear expansion. What happens to the size of a piece of rubber as it is warmed? _____

Part III. Workout: Show all the necessary steps clearly for the following problems /15pt/

1. A horizontal pipe of 10 cm in diameter has a smooth reduction to a pipe of 5 cm in diameter. If the pressure of the water in the larger pipe is 80kPa and the pressure in the smaller is 60kPa, at what rate does the water flow through the smaller pipe? [4pts]

Soln $d_1 = 10\text{cm} = 0.1\text{m}$, $d_2 = 5\text{cm} = 0.05\text{m}$

$$P_1 = 80\text{kPa} = 80 \times 10^3 \text{ Pa}$$

$$P_2 = 60\text{kPa} = 60 \times 10^3 \text{ Pa}$$

$$A_1 = \frac{\pi d_1^2}{4} = 0.00785 \text{ m}^2$$

$$A_2 = \frac{\pi d_2^2}{4} = 0.0019625 \text{ m}^2$$

Using equation of Continuity

$$A_1 V_1 = A_2 V_2 \Rightarrow 0.00785 V_1 = 0.0019625 V_2$$

$$\Rightarrow V_2 = 4 V_1 \quad \text{--- (1)}$$

Applying Bernoulli's equation for horizontal pipe

$$P_1 + \frac{1}{2} \rho V_1^2 = P_2 + \frac{1}{2} \rho V_2^2 \quad \text{--- (2)}$$

$$80\text{kPa} + 500 V_1^2 = 60\text{kPa} + 500 \times 16 V_1^2$$

$$V_1 = \sqrt{\frac{20 \times 10^3}{15 \times 500}} = \sqrt{\frac{200}{75}} = 1.633 \text{ m/s} \quad \text{in larger pipe}$$

in the smaller $V_2 = 6.532 \text{ m/s}$

$$Q = A_1 V_1 = 0.00785 \times 1.631$$

2. In an engine, a piston oscillates with simple harmonic motion so that its position varies according to the expression : $x(t) = 5 \cos\left(2t + \frac{\pi}{6}\right)$

Where x is in meters and t is in seconds. find [4pts]

- (a) The position of the particle, at $t = 0$
- (b) Its time varying velocity
- (c) Its maximum acceleration
- (d) The period

Adama Science and Technology University
School of Applied Natural Science
Department of Applied Physics

General Physics *Final-Exam for Regular Pre-engineering Students for 2020-2021 Ac. yr*

Time Allowed: 2:30hrs and Maximum Mark allotted: 45 %

Full Name _____ ID. No _____ Section: _____

General Direction

- The exam has three parts: Multiple choice, short answer and work out problems.
- Use of **red pen, pencil & cell phone** is not permitted in the exam hall.
- Written manuscripts and printed materials are strictly prohibited in an exam hall.
- You are not allowed to use any extra paper; rather you can use the back page of each paper for your rough work.
- Sharing calculator is strictly prohibited!

Use the following constants wherever applicable: -

density of water (ρ_w) = 1000 kg/m³

specific heat of water (c_w) = 4190 J/kg °C

Earth's gravitational acceleration $g = 10\text{m/s}^2$

Atmospheric pressure $P = 1.01 \times 10^5 \text{pa}$

Specific heat capacity $c_{Al} = 900 \text{ J/kg} \cdot ^\circ\text{C}$

Linear thermal expansion of Al ($\alpha_{Al} = 24 \times 10^{-6} / ^\circ\text{C}$)

Density of aluminum ($\rho_{Al} = 2700 \text{kg/m}^3$)

For Instructors use only!

Multiple choice/15pt/	Short answer/15pt/	Workout /15pt/	Total (45%)



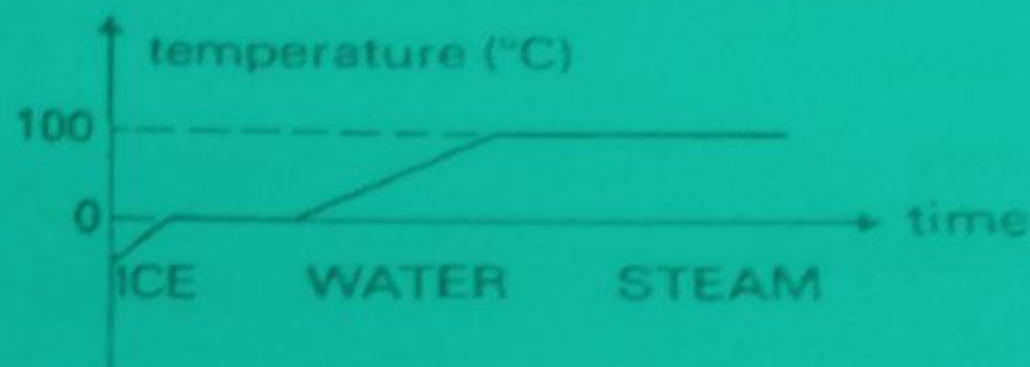
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5. Fluid is a substance that
 - A. Cannot be subjected to shear forces
 - B. Always expands until it fills any container
 - C. Has the same shear stress at a point regardless of its motion
 - D. Cannot remain at rest under action of any shear force
6. Bulk modulus is the ratio of _____.
 - A. Increase of pressure and volumetric strain
 - B. Volumetric stress and increase of pressure
 - C. Volumetric strain and increase of pressure
 - D. Increase of pressure and volumetric stress
7. Which one of the following is NOT TRUE about thermo dynamical process?
 - A. A cyclic process is the one that originates and terminates at the same state
 - B. In adiabatic free expansion, then internal energy of the gas does not change.
 - C. An isothermal process is one that occurs at constant temperature
 - D. The work done on a gas during an isobaric process is given by $W = nRT \ln\left(\frac{V_i}{V_f}\right)$
8. Why does the temperature stop rising at 0°C and 100°C in a temperature verses time graph shown below?



- A. The energy goes not into making the molecules to move faster,
 - B. Work has to be done to hold the molecules together,
 - C. The energy goes to separate the molecules from each other,
 - D. All of the above
9. What mass of steam initially at 130°C is needed to warm 200 g of water in a container from 20.0°C to 50.0°C ? ($C_p = 2010 \text{ J/Kg}^\circ\text{C}$ and $C_w = 4186 \text{ J/Kg}^\circ\text{C}$)

Part I: Write true if the statement is correct and false if the statement is incorrect [1pt each]

1. Oscillation of simple pendulum is an example of linear oscillation.
2. In isolated system, heat does not exchange and work is done on the environment.
3. If the amplitude of SHM is equal to the position then the velocity is constant.
4. If a weight of a body is more than the upward thrust, then the body sinks.

Part II: Choose the best answer from the given alternatives. [1.5pts each]

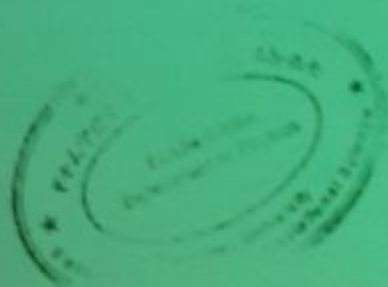
1. Which one of the following is not true?
 - A. All points at the same level in a fluid have the same pressure
 - B. Fluid pressure increases in the depth of the fluid
 - C. Atmospheric pressure decreases with increase in altitude
 - D. Fluid pressure depend on the shape of the container
2. Which property of the fluid offers resistance to deformation under the action of shear force?
 - A. Density
 - B. Viscosity
 - C. Permeability
 - D. Specific gravity
3. Which one of the following is not true about ideal fluids?
 - A. The fluid is viscous
 - B. The flow is steady
 - C. The fluid is incompressible
 - D. The flow is irrotational
 - E. None
4. The radius of a water pipe decreases from 10 to 5 cm. If the average velocity in the wider portion is 4 m/s, find the average velocity in the narrower region.
 - A. 8m/s
 - B. 16m/s
 - C. 4m/s
 - D. 12m/s

Part III: Fill in the blank space

4. The change in length of a rod of length 2m with $\alpha = 10^{-6}/^{\circ}\text{C}$, when it is heated through 5K, is _____.
5. The angular position of a pendulum is represented by the equation $\theta = 0.0320 \cos \omega t$, where θ is in radians and $\omega = 4.43 \text{ rad/s}$. Then the period and length of the pendulum respectively are _____ and _____.
6. The total energy of a particle executing SHM of period 2π seconds is $10.24 \times 10^{-3} \text{ J}$. The displacement of the particle at $\pi/4 \text{ s}$ is $0.08 \sqrt{2} \text{ m}$. Then the mass of particle is _____.
7. The heat absorbed or released when matter melts, changing phase from solid to liquid form at constant temperature is _____.
8. On a day when the temperature reaches 50°F , then the equivalent temperature in degree Celsius is _____.
9. A bimetal plate consists of two materials of different coefficients of thermal expansion. The coefficient of thermal expansion of the top part of the plate is less than the bottom part.
 - (a) If the temperature of the entire plate increases then the plate bends _____.
 - (b) If the temperature of the entire plate decreases, then the plate bends _____.
10. A block of wood weights 4N in air and 3N when immersed in liquid. Then the buoyant force on the block wood is equal to _____.

Part IV: Work out problems (10 pts)

1. In an engine, a piston oscillates with simple harmonic motion so that its position varies according to the expression
 $x(t) = (5.00 \text{ cm}) \cos(2t - \pi/6)$ where x is in centimeters and t is in seconds. At $t = 0$, find
 - a) the position of the piston,
 - b) its velocity
 - c) its acceleration
 - d) the period and
 - e) amplitude of the motion.
2. Calculate the quantity of heat required to raise the temperature of 1 g of ice from -10°C to 110°C .
 $L_v = 2256 \text{ KJ/Kg}$, $L_f = 334 \text{ KJ/Kg}$, $C_v = 2010 \text{ J/Kg}^{\circ}\text{C}$, $C_w = 4186 \text{ J/Kg}^{\circ}\text{C}$ and $C_s = 2090 \text{ J/Kg}^{\circ}\text{C}$





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Part I. Multiple choice questions (1.5pt each)

- Which statement is correct about the pressure under a fluid in a container?
(A) It depends on the volume of the fluid (C) It depends on the density of the fluid
(B) It depends on the shape of the container (D) It is independent of the depth of the fluid
- A table-tennis ball which is spherical in shape has a diameter of 6 cm and average density of 0.4 g/cm^3 . What force is required to hold it completely submerged under water?
(A) 0.68 N downward (C) 1.13 N downward
(B) 0.68 N upward (D) 1.13 N upward
- Lead has a greater density than iron, and both are denser than water. Comparing the buoyant force on the lead object to the buoyant force on the iron object of same volume.
(A) the buoyant force on the lead object is greater than the buoyant force on an iron object
(B) the buoyant force on the lead object is less than the buoyant force on an iron object
(C) the buoyant force on the lead object is equal to the buoyant force on an iron object
(D) None of the above
- Two tubes of different cross-sectional areas are connected together horizontally. Which of the following is a true statement about a fluid flowing through the tubes
(A) The amount of fluid that leaves the wider tube is less than the amount of fluid that enters the narrower tube in the same interval of time.
(B) The speed of the fluid in the narrower tube is greater than the speed of the fluid in the wider tube.
(C) The pressure of the fluid in the narrower tube is greater than that in the wider tube.
(D) The speed of the fluid in the narrower tube is less than the speed of the fluid in the wider tube.
(E) The amount of fluid that leaves the wider tube is greater than the amount of fluid that enters the narrower tube in the same interval of time.
- A garden hose has an inside diameter of 16 cm. The hose can fill a 10 liter bucket in 2 seconds. If the diameter of the nozzle is reduced to 8 cm,
(A) Water exits with a speed twice the speed inside the hose.
(B) A 10 liter bucket can be filled in 10 seconds
(C) A 20 liter bucket can be filled in 40 seconds
(D) Water exits with a speed half the speed inside the hose.

Part II. Give short answer /1.5pt each/

1. A layer of oil with density 800 kg/m^3 floats on top of a volume of fresh water. A block of some material of density 950 kg/m^3 , a uniform cross-sectional area A and height h floats at oil-water interface, as shown in figure below.



(a) What fraction of the block is in water? _____

(b) What fraction of the block is in oil? _____

2. An object is suspended by a cord as shown in *figure 'a'* and the tension in the cord is $T_1 = 10 \text{ N}$. The object is then immersed to the water as shown in *figure 'b'*. In this case the tension in the cord is $T_2 = 6 \text{ N}$.

Determine the density of the object. _____



Fig a

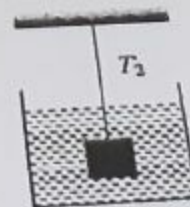


Fig b

3. A small block of wood, of density $0.4 \times 10^3 \text{ kg/m}^3$ is submerged in water at a depth of

6. When a load of 500 kg is hanging from a steel wire of length 3 m and cross sectional area 0.2 cm^2 , the wire stretches beyond its no load length. If the young's modulus for the wire made of steel is $Y = 2 \times 10^{11} \text{ Pa}$, what is the change in length of the wire beyond its no-load length?
- (A) 3.75 mm (B) 2.75 mm (C) 1.75 mm (D) 0.75 mm
7. What is Thermal expansion?
- A) The transfer of thermal energy between materials by the collisions of particles
 B) Is a decrease in a materials size when temperature decreases
 C) An increase in a materials size when temperature increases
 D) All
8. Which of the following statements is false?
- A) Gaps need to be left in lengths of railway lines to prevent buckling in hot weather
 B) Bimetallic strips are used in thermostats, a thermostat being a temperature- operated switch
 C) As the temperature of water is decreased from 4°C to 0°C contraction occurs
 D) A change of temperature of 15°C is equivalent to a change of temperature of 15 K
9. If the coefficient of linear expansion is A, the coefficient of superficial expansion is B and the coefficient of cubic expansion is C, which of the following is false?
- A. $C = 3A$ B. $A = B/2$ C. $B = 3C/2$ D. $A = C/3$
10. The temperature of land rises more quickly than that of the sea because the specific heat of soil is
- A) more than water C) less than water
 B) equal to water D) neutral

Part I. Multiple choice questions (1.5pt each)

- Which statement is correct about the pressure under a fluid in a container?
(A) It depends on the volume of the fluid (C) It depends on the density of the fluid
(B) It depends on the shape of the container (D) It is independent of the depth of the fluid
- A table-tennis ball which is spherical in shape has a diameter of 6 cm and average density of 0.4 g/cm^3 . What force is required to hold it completely submerged under water?
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(A) Water exits with a speed twice the speed inside the hose.
(B) A 10 liter bucket can be filled in 10 seconds
(C) A 20 liter bucket can be filled in 40 seconds
(D) Water exits with a speed half the speed inside the hose.

4. A glass contains 0.25kg of Omni-Cola (mostly water) initially at 25°C . How much ice, initially at -20°C , must you add to obtain a final temperature of 0°C with all the ice melted? Neglect the heat capacity of the glass. ($c_{\text{water}} = 4190 \text{ J / kg }^{\circ}\text{C}$ and $c_{\text{ice}} = 2100 \text{ J/kg}^{\circ}\text{C}$, and $L_f = 3.34 \times 10^5 \text{ J/kg}$) [3pts]

3. A 1kg block of aluminum is heated at atmospheric pressure so that its temperature increases from 22°C to 40°C . Find [4pts]

- (a) The work done on the aluminum,
- (b) The energy added to it by heat, and
- (c) The change in its internal energy

2. In an engine, a piston oscillates with simple harmonic motion so that its position varies according to the expression : $x(t) = 5 \cos \left(2t + \frac{\pi}{6} \right)$

Where x is in meters and t is in seconds. find [4pts]

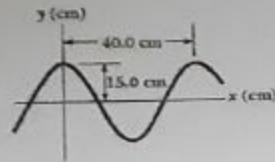
- (a) The position of the particle, at $t = 0$
- (b) Its time varying velocity
- (c) Its maximum acceleration
- (d) The period

Part III. Workout: Show all the necessary steps clearly for the following problems /15pt/

1. A horizontal pipe of 10 cm in diameter has a smooth reduction to a pipe of 5 cm in diameter. If the pressure of the water in the larger pipe is 80kPa and the pressure in the smaller is 60kPa, at what rate does the water flow through the smaller pipe? [4pts]

7. The condition of resonance occurs when the driving frequency becomes equal to _____ of the oscillator.

8. A sinusoidal wave traveling in the positive x-direction has a wavelength of 40cm, and a frequency of 8Hz. The vertical position of an element of the medium at $t = 0$ and $x = 0$ is also 15cm as shown in figure below. The period T & linear speed of the wave are _____ and _____ respectively.



9. A car with a mass of 1300 kg is constructed so that its frame is supported by four springs. Each spring has a force constant of 20,000 N/m. Two people riding in the car have a combined mass of 160 kg. Find the frequency of vibration of the car after it is driven in the road and the car oscillates vertically. _____

10. Rubber has a negative average coefficient of linear expansion. What happens to the size of a piece of rubber as it is warmed? _____



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7. Lead has a greater density than iron, and both are denser than water which one is true about the buoyant force on a lead and the buoyant force on an iron object of the same volume?

- A. greater than ☒ B. the same ☐ C. less than D. none

8. A balloon lifting in air follows the following principle:

- A. Law of gravitation C. principle of buoyancy
B. Archimedes' principle D. all are correct

9. A horizontal pipe narrows from a diameter of 10 cm to 5 cm. For a non-viscous liquid flowing without turbulence from the larger diameter to the smaller,

- ☒ A. The velocity and pressure both increases
B. The velocity increases and the pressure decreases
C. The velocity decreases and the pressure increases
D. The velocity and pressure both decreases

$$F_b = S \cdot g \cdot V_0$$

$$P_1 = S + g \left(\frac{m}{S_1} \right)$$

$$F_{b2} = S + g \left(\frac{m}{S_2} \right)$$

$$P = F$$

$$\frac{1}{b}$$

10. Heat is transmitted from a place of higher temperature to a place of lower temperature by means of particles with migration from one place to another

- A. Conduction C. Radiation
☒ B. Convection D. Direct burning

$$F_{b1} = S + g \cdot V_0$$

$$P_{b2} = S + g \cdot V_0$$

11. Consider the following pairs of materials. Which pair represents two materials, one of which is twice as hot as the other?

- ☒ A. Boiling water at 100°C, a glass of water at 50°C
B. Boiling water at 100°C, frozen methane at -50°C
C. An ice cube at -20°C, flames from a circus fire-eater at 233°C
D. None of those pairs

12. Which one of the following is incorrect in **isobaric process**?

- ☒ A. The Volume of the system stay constant, and consequently work done is zero
B. Expansion and compression occur at constant pressure
C. Any work done by the system results in a decreasing a volume
D. The work done in the pressure-Volume graph is the area under the PV graph

13. An amount of energy is added to ice, raising its temperature from -10 °C to -5 °C. A larger amount of energy is added to the same mass of water, raising its temperature from 15 °C to 20 °C. From these results, what would you conclude?

$$-5 + 10 = 5$$

Name: _____ ID No. _____ Sec. _____

Part III: Choose the correct answer among the given alternatives (1.5 pts each)

1. If a water bottle is knocked off a desk, when does the bottle have the **MOST** kinetic energy?
A. At the top of the fall ☒ C. At the bottom of the fall
B. In the middle of the fall D. The bottle has the equal kinetic energy in all level
2. Two particles colliding in perfectly inelastic collision which one of the following is NOT correct about two objects?
A) They have common velocity after collision
B) They stick together after collision
☒ C) They have the same kinetic energy before and after collision
D) They have the same momentum before and after collision
3. Two-point masses 3 kg and 5 kg are at 4 m and 8 m from the origin on X-axis, respectively. Locate the position of center of mass of the two-point masses (i) from the origin and (ii) from 3 kg mass, respectively.
A. (8.5 and 7.5) m C. (4.5 and 3.5) m
☒ B. (6.5 and 2.5) m D. (14.5 and 11.5) m
4. Bernoulli's equation is based on
A. Conservation of momentum ☒ C. Conservation of energy
B. Conservation of mass D. Mass-energy equivalence
5. If the net work done by external forces on a particle is zero, which of the following statements about the particle must be true?
A. Its velocity is zero C. Its velocity is unchanged
B. Its velocity is decreased D. Its speed is unchanged
$$W = \int \vec{F} \cdot d\vec{s} = F(vt) = F(0)(t)$$
6. A Hydraulic press has a ram of 30cm diameter and a plunger of 2cm diameter. It is used for lifting a weight of 35 kN. Find the force required at the plunger.
A. 233.3 kN B. 311.1 kN C. 466.6 kN ☒ D. 0.1555 kN

Name: _____

ID No. _____

Sec. _____

Part I: Say True if the statement is correct and False if the statement is incorrect (1pt each)

1. If two or more particles in an isolated system interact to each other, the total momentum of the system remains constant.
2. The equation of continuity is derived from the law of conservation of mass of compressible fluid.
3. The center of mass of an object lies closer to the more massive particle.
4. In isothermal processes the work done is equal to the amount of heat energy absorbed by a system.
5. The pressure of a fluid decreases as increase in fluid depth.
6. The change in internal energy of a cyclic process is zero.

Part II: Match the following from column "B" to column "A" (1 pt each)

A

1 Stress ρ

2 Shear modulus η

3 Strain ϵ

4 Bulk modulus c

5 Young modulus λ

B

A. Measures the resistance of a solid to a change in its length

B. Measures the resistance to motion of the planes within a solid parallel to each other

C. Measures the resistance of solids or liquids to changes in their volume

D. The external force acting on an object per unit cross-sectional area.

E. A measure of the degree of deformation.

F. The fractional change in length of an object under a tensile stress.

G. The ratio of horizontal distance that the sheared face moves to the height of the object

- A. Overcoming the latent heat of fusion of ice requires an input of energy.
 B. The latent heat of fusion of ice delivers some energy to the system.
 C. The specific heat of ice is less than that of water.
 D. The specific heat of ice is greater than that of water.
 E. More information is needed to draw any conclusion.

$$Q = mc\Delta T$$

$$C_{ice} = \frac{Q}{m\Delta T}$$

$$C_{ice} = \frac{2200}{m\Delta T}$$

14. A 50.0-g sample of copper is at 25.0 °C. If 1200 J of energy is added to it by heat, what is the final temperature of the copper?

A. 87.0 °C B. 87.0 °K C. 87.0 °F D. All

$$C_{copper} = \frac{Q}{m\Delta T}$$

15. Today's temperature of Wolkite university is 25 °C, what is its temperature in degree Fahrenheit and degree Kelvin respectively.

A. 13 and 248.15 B. 77 and 298.15 C. 46 and 298.15 D. -18 and 248.15

$$C_{ice} = \frac{Q}{m\Delta T}$$

16. Which of the following is not true about first law of thermodynamics?

- A. For an isolated system internal energy is constant.
 B. For a system interact with environment but thermally isolate, change internal energy is equal to work done by the environment on the system.
 C. For a system interact with environment change internal energy is the sum of heat added and work done on the environment.
 D. For a system of only thermally interacting, heat added to the system is equal to change in internal energy.

$$273$$

$$273$$

$$273$$

$$45$$

$$32$$

$$27$$

Part IV: Briefly state the following questions (5pts)

1. Briefly state the zeroth and First law of thermodynamics? (2 pts)

2. State the following terms briefly

- i) Work-energy theorem (1.5pts)
 ii) Archimedes principle (1.5pts)

$$Q = mc\Delta T$$

$$Q = mc(\Delta T - T_i)$$

$$1200 = 0.05 \times (327 - T_i) \Delta T$$

Part V: Solve the following problems by showing all the necessary steps (10 Pts each)

- A gas is taken through the cyclic process described in Figure 1. (2.5 pts each)

- a. Find the net energy transferred to the system by heat during one complete cycle.

$$5 \quad \Delta T = 1220$$

$$1935 \times 10^2$$

$$240$$

$$147$$

$$C = \frac{Q}{m\Delta T}$$

$$C = \frac{1200}{5 \times 10^{-2} \times 67}$$

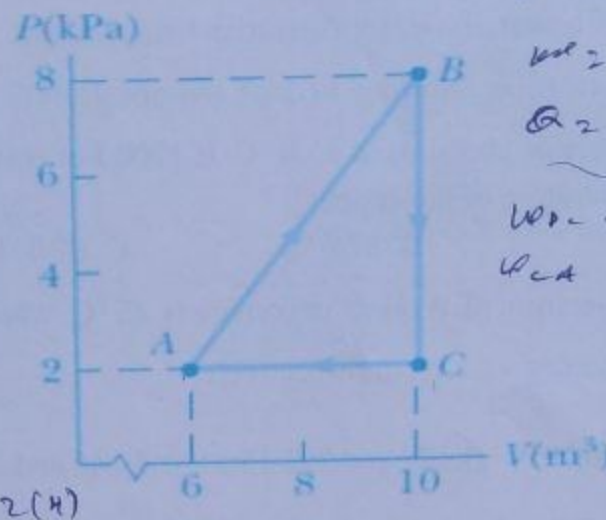
$$1200$$

$$333$$

- b. If the cycle is reversed, the process follows the path **ACBA**, what is the net energy input per cycle by heat?

$$\frac{P \Delta V}{RT} = \frac{nRT}{RT}$$

$$n =$$



$$W = A = \frac{1}{2}(6)(4) = 12$$

$$W = -12 \text{ J} = -1.2 \times 10^4 \text{ J}$$

$$Q = -W = 1.2 \times 10^4 \text{ J}$$

$$W_{AB} = 0 \quad W_{BC} = -2$$

$$W_{CA} = 8$$

$$W = -nRT \ln\left(\frac{V_f}{V_i}\right)$$

$$W = -1.2 \times 10^4 \text{ J}$$

2. An object of mass 2kg is released from a top of inclined plane of 30° and height 6m. The coefficient of kinetic friction of the surface is 0.15. (5 pts)

$$m = 2 \text{ kg}$$

$$h = 6 \text{ m}$$

$$\mu_k = 0.15$$

$$W_f = (\mu_k N)(S)$$

$$W_f = 0.15(20)(\sqrt{3}/2)(6)$$

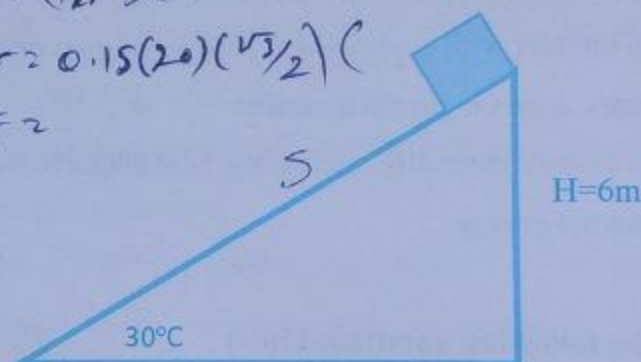
$$W_f =$$

$$A = 12 \quad K = 4(2) = 8$$

$$= 20$$

$$N = mg \cos \theta$$

$$W_f$$



- A. What is the work done against friction force when the object reaches the bottom?
 B. What is the kinetic energy of the object when it reaches the bottom?
 C. What is the speed of the object when it reaches the bottom?

$$KE = \frac{1}{2}mv^2$$

Answer sheet

Part I

1. True
2. False
3. true
4. True
5. False
6. True

6

Part II

1. D
2. B
3. E
4. C
5. A

5

Part III

1	<u>C</u>	9	<u>A</u>
2	<u>C</u>	10	<u>B</u>
3	<u>B</u>	11	<u>A</u>
4	<u>C</u>	12	<u>A</u>
5	<u>A</u>	13	<u>C</u>
6	<u>D</u>	14	<u>A</u>
7	<u>B</u>	15	<u>B</u>
8	<u>B</u>	16	<u>C</u>

$$12 \times \frac{3}{2} = \frac{18}{2} = 9$$

Sum

Sum

Sum



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Full Name: _____

Instructor's Name: _____

ID: _____

Section: _____

For Instructors' Use

Part I (40%)	Part II (10%)	Total (50%)

Useful Constant: $g = 10 \text{ m/s}^2$, $\rho_w = 1000 \text{ kg/m}^3$

CALCULATOR AND MOBILE IS NOT ALLOWED AT ALL

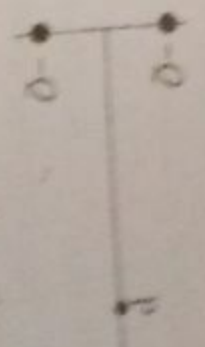
Part I: Multiple Choices (40%)

Instruction: Choose the best answer and write your choice in Capital letter on the space provided for each question.

1. A hydraulic press has one piston of diameter 2.0 cm and the other piston of diameter 8.0 cm. What force must be applied to the smaller piston to obtain a force of 1600 N at the larger piston?
A. 400 N B. 100 N C. 25 N D. 6.25 N
2. A sample of ideal gas has an internal energy U and is then compressed to one-half of its original volume while the temperature stays the same. What is the new internal energy of the ideal gas?
A. $2U$ B. $\frac{1}{2}U$ C. $\frac{1}{4}U$ D. U
3. A weight suspended from an ideal spring undergoes a simple harmonic motion with a period T . If the amplitude of the oscillation is doubled, the period will be: A. $T/2$ B. $1.5T$ C. $2T$ D. T
4. A sample of an unknown material weighs 400 N in air and 300 N when submerged in a liquid with a density of $0.5 \times 10^3 \text{ kg/m}^3$. What is the density of the material?
A. $3.0 \times 10^3 \text{ kg/m}^3$ B. $0.6 \times 10^3 \text{ kg/m}^3$ C. $2.0 \times 10^3 \text{ kg/m}^3$ D. $0.25 \times 10^3 \text{ kg/m}^3$
5. Which of the following statement is correct about the pressure of a fluid with in a container?
A. It depends on the density of the fluid C. It depends on the volume of the fluid
B. It is independent of the depth of the fluid D. It depends on the shape of the container
6. Salt water is more dense than fresh water. A ship floats in both fresh water and salt water. Compared to the fresh water, the amount of water displaced in the salt water is
A. Cannot be determined from the information given.
B. the same. C. more. D. less
7. The pressure exerted on the ground by a man is greatest when
A. he stands with both feet flat on the ground C. he stands on the toes of one foot
B. he lies down on the ground D. all of the above yield the same pressure
8. A solid cube made of wood has sides of length 0.1 m. The mass of the cube is 0.3 kg. The specific gravity of the wood is _____. A. 0.3 B. 3.3 C. 1.3 D. 0.6

1
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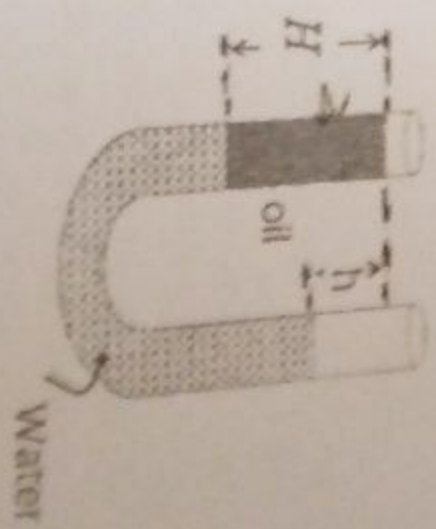
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9. A system undergoes an adiabatic process in which its internal energy increases by 20 J. Which of the following statements is true?
- A. 20 J of work was done by the system
B. 20 J of work was done on the system
C. the system received 20 J of energy as heat
D. the system lost 20 J of energy as heat
10. A 2m-long rod of cross section $4 \times 10^{-6} \text{ m}^2$ was found to stretch with 200N by a force of 120 N. The Young's modulus of the rod is _____
- A. $6 \times 10^9 \text{ N/m}^2$
B. $4.5 \times 10^9 \text{ N/m}^2$
C. $1.5 \times 10^9 \text{ N/m}^2$
D. $1 \times 10^9 \text{ N/m}^2$
11. As a liquid goes through a phase change from liquid to solid, heat is released and the temperature _____
- A. remain the same
B. fluctuates
C. increases
D. decreases
12. If the ice-point and the steam-point temperatures on an unknown scale X are 80°X and 180°X , respectively, what is the reading in $^\circ\text{X}$ for a temperature of 40°C ?
- A. 140°X
B. 120°X
C. 80°X
D. 60°X
13. For simple harmonic motion, there must be a restoring force proportional to the _____
- A. Amplitude
B. frequency
C. velocity
D. displacement
14. A magnetic field is directed perpendicular to the plane of a circular coil of area 0.2 m^2 and has 250 turns. The magnetic field increase from 0.01 T to 0.06 T in a time interval of 0.25 s . What is the average electromotive force induced in the coil?
- A. 10V
B. 0.1V
C. 2.5V
D. 0.04V
15. Which one of the following waves do not require material medium for propagation?
- A. water wave
B. sound wave
C. Electromagnetic wave
D. wave on a string
16. The diagram shows two negative charges $-Q$ each placed on the y axis at equal distances from the origin. The direction of the electric field at point P on the x axis is _____.
- 
- A. to the left
B. to the right
C. downward
D. upward
17. A resistor in a circuit dissipates energy at a rate of 1 W. If the voltage across the resistor is doubled, what will be the new rate of energy dissipation?
- A. 0.5W
B. 4W
C. 0.25W
D. 2W
18. How much work is done on a proton to move it from the negative plate to a positive plate 2m away if the electric field between the plates is 200 N/C ?
- A. $1.6 \times 10^{-17} \text{ J}$
B. $3.2 \times 10^{-17} \text{ J}$
C. $4.8 \times 10^{-17} \text{ J}$
D. $6.4 \times 10^{-17} \text{ J}$
19. A proton is moving to the right edge of this page in a uniform magnetic field directed upward. The direction of the magnetic force is _____
- A. out of this page
B. to the top of this page
C. into this page
D. to the bottom of this page

29. A simple pendulum of length l has bob of mass m and oscillates with a frequency f . If the pendulum is replaced by another one of length $l/4$, the frequency of oscillation will be ____.
- A. f B. $f/4$ C. $2f$ D. $f/2$

Part II: Short Answer.
Write the precise answer on the space provided on the question paper. (2pts each)

21. An unknown resistor R and a 12Ω resistor are connected to produce an equivalent resistance of 3.0Ω .
a) The type of connection is _____. b) The value of R is _____ Ω .
22. Two small charged objects repel each other with a force F_0 when separated by a distance d . If the charge on each object is reduced to half of their initial value and the distance between them is increased to $2d$, the new force in terms of F_0 becomes _____.
23. An object oscillates with simple harmonic motion along the x axis. Its position varies with time according to the equation, $x = \left(\frac{5}{\pi}\right)m \left(\sin 4\pi t - \pi/4\right)$ where t is in seconds and the angles in the parentheses are in radians and m is meter.
a) The period of the oscillation is _____ s.
b) The maximum speed of the wave is _____ m/s.
24. Metal A has a coefficient of area expansion that is three times the coefficient of area expansion of metal B. The coefficient of volume expansion of metal B in terms of coefficient of volume expansion of metal A, $\gamma_B = \gamma_A$.
25. Water and oil in a U tube become equilibrium as shown. The height difference h as a function of H , density of water ρ_w and density of oil ρ_o is $h = \frac{\rho_w}{\rho_o} H$.
- (Consider the two ends of the tube are open and air pressure is the same on the two ends)



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INSTRUCTIONS:
PART-I: Choose the Best Answer for the Following Questions and
Write the Letter of your Choice in the Answer sheet Provided (1.5 pt. each)

1. Which one of the following is not an example of mechanical wave?
A) Sound wave
B) Water wave
C) Light wave
D) P-type earth quake wave
2. is a type of wave in which the motion of particles of the disturbed medium are parallel to the wave motion
A) Transverse Wave
B) Longitudinal Wave
C) Both A and B
D) Neither A nor B
3. Which one of the following is the characteristics of a wave?
A) Reflection of wave
B) Refraction of wave
C) Interference of wave
D) All
4. In Doppler effect, i.e the frequency of sound heard by an observer is
A) decreases as the source of sound is approaching to the observer
B) remain constant as the source of sound is approaching to the observer
C) increases as the source of sound is receding from the observer
D) increases as the source of sound is approaching to the observer
5. The physics of electric charges at rest is —
A) Current electricity
B) Electrostatics
C) Electronics
D) Magnetism
6. As the separating distance between two equal point charges is halved, then the magnitude of the electrostatic force will be —
A) doubled
B) halved
C) one fourth
D) four times

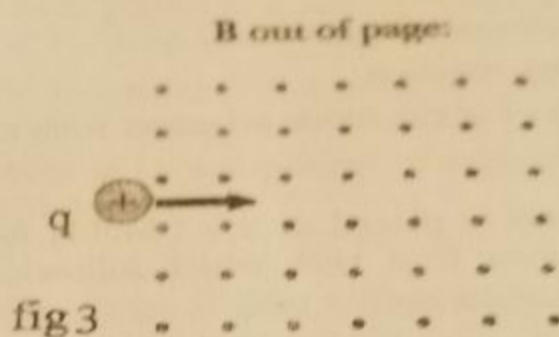
7. How much heat is required to change 2g of ice at 0°C to water at 40°C , Latent heat of fusion of ice ($L_f = 80\text{cal/g}$) and specific heat capacity of water $c_w = 1\text{cal/g}^{\circ}\text{C}$
- A) 160 Cal C) 100 Cal
B) 40 Cal D) 200 Cal
8. Two bodies are in thermal equilibrium if they have
- A) the same pressure
B) different temperature
C) the same chemical reaction
D) the same temperature
9. At what temperature are the Celsius and Fahrenheit scale equal?
- A) -50°C C) -30°C
B) -40°C D) -20°C
10. If the ice-point and the steam-point temperatures on a Belay scale are 20°B and 150°B , respectively, then what is the reading in $^{\circ}\text{B}$ for a temperature of 15°C ?
- A) 10°B C) 40°B
B) 30°B D) 50°B
11. Which one of the following is the correct relation of coefficients of thermal expansion?
- A) $\beta = 2\alpha$ C) $3\beta = 2\gamma$
B) $\gamma = 3\alpha$ D) All
12. The electric field strength at the mid point of two equal and similar charges(q) is—
(let the distance between a charge and the midpoint be r)
- A) $\frac{kq}{r^2}$ C) 1
B) 0 D) $\frac{kq}{r}$
13. Which one of the following is not a way to induce emf?
- A) Changing the magnitude of the magnetic field with time
B) Changing the area of the loop in the field
C) Changing the orientation of the loop in the field
D) None

14. _____ is a type of thermodynamic process that occurs at constant temperature.

- A) Isochoric process
- B) Adiabatic process
- C) Isobaric process
- D) Isothermal process

15. A positive charge moving with constant velocity $\vec{v}$ enters region of a uniform magnetic field pointing out of the page (as depicted fig3). what is the direction of the magnetic force on the charge?

- A) To the Right
- B) To the top of the page
- C) To the Left
- D) To the bottom of the page

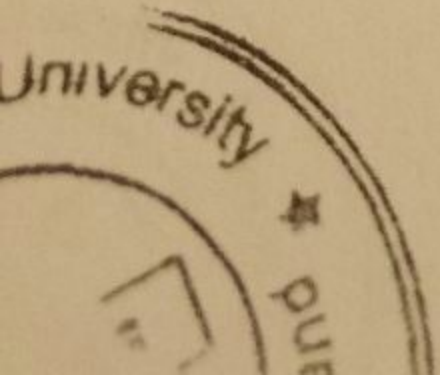


16. As the temperature of a copper wire increases, its resistance—

- A) increases
- B) decreases
- C) remains constant
- D) neither increases nor decreases

17. Which one of the following is not a thermodynamic variable?

- A) Temperature
- B) Volume
- C) Pressure
- D) None



18. The total resistance of two resistors when they are connected in series and in parallel separately are 9 ohm and 2 ohm respectively. the resistances of resistors are—
- A) 5 ohm and 4 ohm B) 3 ohm and 6 ohm
C) 1 ohm and 8 ohm D) 2 ohm and 7 ohm
19. Which one of the following is true about simple harmonic motion(SHM)?
- A) The magnitude of restoring force is directly proportional to the displacement
B) The direction of restoring force is opposite to the displacement
C) The speed is maximum at equilibrium position
D) All
20. The period of a simple harmonic motion(SHM) does not depend on
- A) the amplitude of oscillation
B) the spring constant
C) the mass of the block attached with a spring
D) B and C
21. When an object is placed at $2f_1$ (double focal length) on one side of the convex lens, then which following is true about the nature of the image?
- A) Virtual, erect, diminished
B) Real, inverted, diminished
C) Real, inverted, enlarged
D) Real, inverted, the same size of the object.
22. Which one of the following is an application Faraday's Law?
- A) Generator
B) Transformer
C) Both A and B
D) None

23. For a circular loop of wire (depicted in fig.2) being dropped towards a wire carrying a current to the left. The direction of the induced current in the loop of wire is—
- A) Clockwise C) Zero
B) Counterclockwise D) Impossible to determine

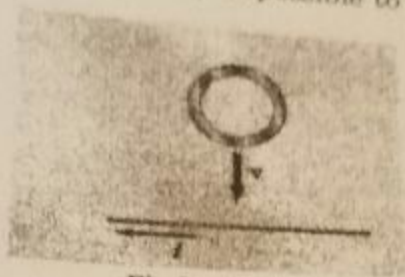


Fig-2

24. A loop of wire is placed in a uniform magnetic field. For what orientation of the loop is the magnetic flux maximum?
- A) When the magnetic field is perpendicular to the loop of the wire
B) When there is no component of magnetic field perpendicular to the loop
C) A and B D) None

PART-II: Write True if the statement is correct or False if the statement is not correct (1 pt each)

1. In an adiabatic process, energy is transferred by heat between the system and its surroundings.
2. All waves carry energy and momentum.
3. Electromagnetic waves can propagate through a material medium and vacuum.
4. Kirchhoff's first rule (or the junction rule) is based on conservation of energy.
5. The image formed by concave lens is always virtual, erect and diminished.
6. When two bodies are in thermal equilibrium, heat can flow from one body to another.
7. Magnet is not the only sources of magnetic field.



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- True or False: Write True if the statement is correct or False if it is incorrect on the space provided (1 pint each)

- _____ 1. The work done by a conservative force along a closed path is always non-zero.
- _____ 2. Both momentum and kinetic energy are conserved in perfectly- inelastic collision.
- _____ 3. Mechanical energy may not be conserved in an isolated spring mass system.
- _____ 4. Uniformly accelerated motion is possible only on a straight path.

Part II - Multiple Choice: Choose the best answer and write the letter of your choice on the space provided (2 pints each)

- _____ 1. One of the following is not true according to Newton's Laws of motion?
a) to every action there is equal and opposite reaction. b) The body is said to be in equilibrium if it is either at rest or in uniform motion.
c) Freely falling object is in gravitational equilibrium
d) none
- _____ 2. Which of the following is not true about uniform circular motion?
a) its acceleration is always radial b) its tangential speed is constant c) It has uniform velocity d) none
e) none
- _____ 3. Object A is fired horizontally with a very large speed while a second object B is released from the same elevation at the same time. Which one of the following is not true about the motion of objects?
a) Object B reaches the ground first b) Object A hits the ground with larger speed
c) Both of them reach the ground equally d) Object A covers larger horizontal range d) none
- _____ 4. Which one of the following motion has non-uniform acceleration?
a) Projectile motion b) uniform circular motion c) uniform motion d) none
- _____ 5. For a given projectile motion, the maximum possible range occurs for the launching angle of
a) 45° b) 60° c) 30° d) 72° e) none
- _____ 6. If a constant force of 100 N is applied for 2 minutes to a 20 kg mass which is initially at rest, the speed acquired and the distance traveled respectively are:
a) 60 m/s and 3.6 km b) 600 m/sec and 36 km c) 600 m/s and 72 km d) none

Part III- Short Answer: Give short and brief explanation for the following questions.

1. State and explain the work energy theorem

(2 pts)

2. What is the center of mass of the system?

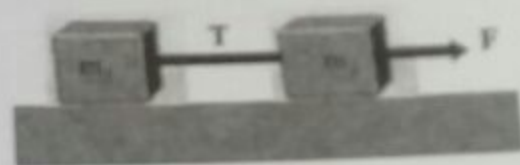
(2 pts)

Part III- Short Answer: Show all the necessary steps with clear hand writing

1. Upstretched rope of negligible mass connects two blocks of masses $m_1 = 12 \text{ kg}$, $m_2 = 18 \text{ kg}$ that are dragged by force $F = 60 \text{ N}$ along a smooth horizontal surface as shown below. Calculate:

a) the acceleration of the two masses

(3pts)



b) the tension T in the rope

(3pts)

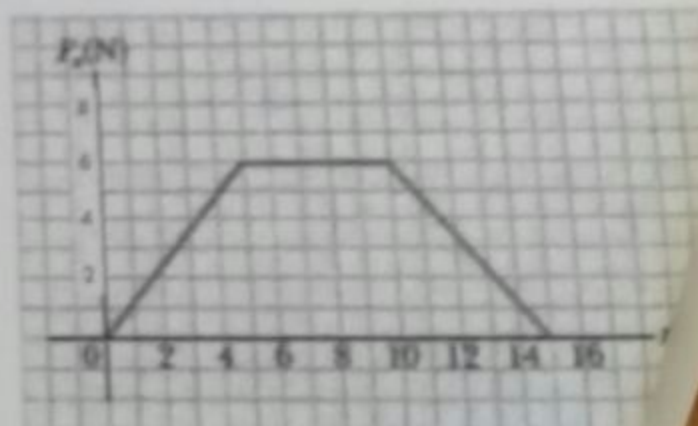
2. A particle of mass 6 kg is subject to a force that varies with position as shown in figure below. Find the work done by the force on the particle as it moves

a) From $x = 0$ to $x = 5 \text{ m}$

(2pts)

b) From $x = 5$ to $x = 10 \text{ m}$

(2pts)



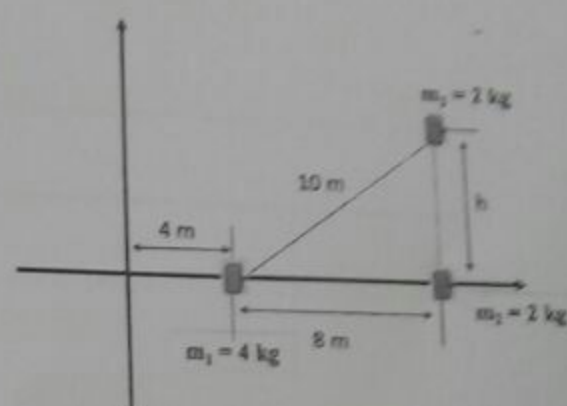
c) From $x = 10 \text{ m}$ to $x = 15 \text{ m}$

(2pts)

d) Find the speed of the particle at 5 m

(4pts)

3. Three particles are located at corner of a right angle triangle as shown in the figure below. Find the center of mass of the system (6pts)



4. A block of mass 680 kg attached to horizontal spring whose spring constant is 65 N/m . the block pulled to a distance of 11 cm from the mean position (Amplitude) and released from rest. Calculate :
- i. Angular frequency (2pts)

ii. Time period (2pts)

iii. Maximum speed (2pts)

iv. Total mechanical energy of SHM system (2pts)



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Part I: Say 'True' if the statement is true or 'False' if not (1 point each)

1. The work done by the force of gravity is the same in raising an object through a height h vertically, as raising it through a rampage of certain inclination to the same height.
2. Elastic collision is the type of collision in which energy and linear momentum are conserved.
3. A body of mass " m " moving around the sun with a speed " v " has maximum potential energy when it is closer to the sun.
4. Two objects are in thermal equilibrium with each other if they do not exchange energy when in thermal contact.
5. Elastic materials are materials that regain their original shape and size.
6. If the weight of an object is greater than the upthrust force acting on it, then the object would sink.
7. Physical contact is necessary for heat to flow from one object to another.
8. A phase change is characterized by a change in thermal energy without a change in temperature.
9. Fluid pressure does not depend on the shape of the container.
10. Solids expand better than liquids.

Part II: Choose the Best Answer (1.5 point each)

1. Two objects with unequal masses are released from rest from the same height. They slide without friction down a slope and then encounter a rough horizontal region. The coefficient of kinetic friction in the rough region is the same for both masses. Which of the following statements is/are true?
A). both masses start out with the same gravitational potential energy
B). both objects have the same speed when they reach the base of the slope
C). both masses have the same kinetic energy at the bottom of the slope

11. Both masses travel the same distance on the rough horizontal surface before stopping.
A horizontal cylindrical pipe has a part with a diameter half that of the rest of the pipe. If v is the speed of the fluid in the wider section of the pipe, then the speed of the fluid in the narrower section is _____.

- A) $v/4$ B) $v/2$ C) $2v$ D) $4v$

3. Which of the following is not true about fluid pressure?

- A) All points at same level in a fluid have same pressure
B) Fluid pressure depends on the shape of the container
C) Fluid pressure increases with increase in the depth of the fluid
D) Fluid pressure doesn't depend on the shape of the container

$$\frac{1}{A} v = \frac{1}{A} v$$

4. The amount of heat energy required to raise the temperature of a substance by one degree Celsius is _____.

- A) Specific heat capacity B) Heat capacity C) Latent heat D) None

5. One of the Following is not scale of Temperature,

- A) Fahrenheit B) Celsius C) Kelvin D) Thermometer

6. Suppose you find yourself teetering on the edge of a cliff or the roof of a tall building with a heavy physics book in your hand. As you are about to fall, you suddenly remember how physics can save you! What law did you remember?

- A) The law of inertia B) The law of gravitation
C) The law of conservation of linear momentum D) The law of conservation of energy

7. Change in momentum a system is a result of _____.

- A) acceleration of the system B) net external force on the system
C) an impulse D) all of the above

8. The unit of strain is _____.

- A) Pascal B) Metre C) Newton meter D) None of these.

9. In studying fluids, as pressure increases the speed of the fluids decreases and vice versa.

This statement is all about:

- A) Archimedes' Principle B) Pascal's principle
C) Equation of Continuity D) Bernoulli's principle

10. Which one of the following statements is incorrect about heat transfer mechanism?

- A) Unlike that of conduction and convection, there is no molecular involvement in the transfer of heat through radiation.

$$modu = \frac{L_0}{\Delta L} \mu$$

2

$$\frac{L_0}{\Delta L} = \mu$$

$$\frac{L_0}{\Delta L} = \mu$$

- B) When we sit beside a fire place, the warmth that we feel reaches us through radiation.
 C) Conduction is one way of heat transfer by which heat transfer occurs by means of successive collisions between neighboring particles.
 D) During convection, thermal energy passes from one molecule to the next, while each individual molecule remains at its original position.

Part III: Give only short answer for the following questions (2 points each)

1. What do we mean by an 'isolated system' in physics?
2. When a particle rotates in a circle, a force acts on it, directed towards the center of rotation. Why is it that this force does no work on the particle?
3. Two solid spheres made of lead and aluminum with densities (ρ_{Pb} , ρ_{Al}) respectively have the same mass. What is the ratio of their radii in terms their densities.
4. Why water is very effective as an extinguisher agent?
5. State the first law of thermodynamics.

Part IV: Workout (Show all the necessary steps clearly) (15 points)

1. An older model car accelerates from rest to speed v in 10 seconds. A newer, more powerful car accelerates from rest to $2v$ in the same time period. What is the ratio of the power of the newer car to that of the older car?
2. An archer shoots an arrow toward a target that is sliding toward her with a speed of 2.50 m/s on a smooth surface. The 22.5-g arrow is shot with a speed of 35.0 m/s and passes through the 300-g target, which is stopped by the impact. What is the speed of the arrow after passing through the target?
3. Water circulates throughout a house in a hot water heating system. If the water is pumped at a speed of 0.5 m/s through a 4.0 cm diameter pipe in the basement under the pressure of $3.03 \times 10^5 \text{ Pa}$, what will be the velocity and pressure in a 2.6 cm diameter pipe on the second floor 5.0 m above
4. A segment of steel rail road track has a length of 40 m when the temperature is 273 K . What will be the length when the temperature is raised to 313 K ? (Take $\alpha = 11 \times 10^{-6} / ^\circ\text{C}$).